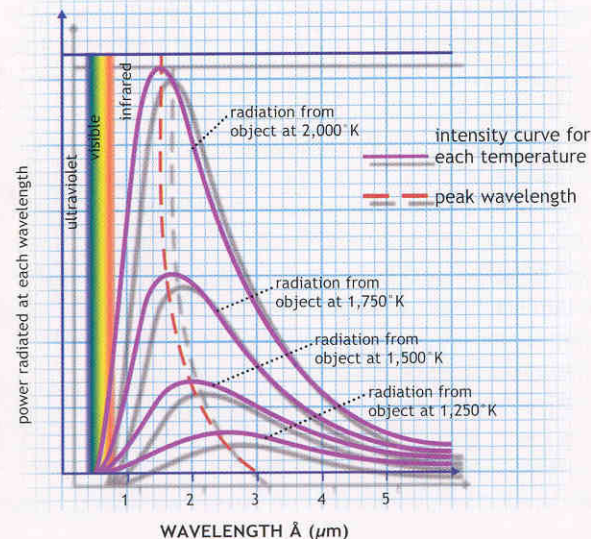




red-hot
A glass blower can tell the temperature of the molten glass, and therefore its physical properties, from its color.

black body curves
The wavelength at which the intensity of the radiation from a hot object peaks is related to the temperature of the object. These "black body curves" all peak in the infrared region of the spectrum in the form of "hills," and produce little radiation in the visible spectrum.



emitted at each wavelength depends on the temperature of the object. This is true in everyday life, and we all know that an object like a radiator may radiate infrared heat without glowing, and that a red-hot lump of iron is cooler than a white-hot lump of iron. Many objects radiate very much like a black body, including the Sun. It is possible to measure the surface temperature of the Sun (about 6,000°K) simply from its color (yellow-orange) and by assuming that it is a black body radiator.

By the 1890s, experiments had shown exactly how the radiation of a black body is related to its temperature. When the electromagnetic spectrum of this radiation is plotted as a graph (the black body curve), it shows a smooth peak, like a hill. At a particular temperature, the peak is always in the same part of the spectrum, that is, at the same wavelength. But, as the black body gets hotter, the peak moves to shorter wavelengths (from infrared, to red, to orange, to blue, and so on).

But there is very little radiation on either side of the peak.

This posed a profound puzzle that could not be explained by classical physics.

It goes like this. If

electromagnetic waves are treated in the same way (mathematically speaking) as waves on the ocean or on a violin string, and if energy (heat) is put in, the intensity of the radiation produced should be proportional to the frequency of the radiation. The higher the frequency (which means the shorter the wavelength), the more radiation there should be—at any temperature. Most of the energy emitted should be in the ultraviolet region and there should be no "hill" within the graph, but there is. This "ultraviolet catastrophe" spelt the beginning of the end of classical physics as a complete explanation of the physical world.

Planck's constant

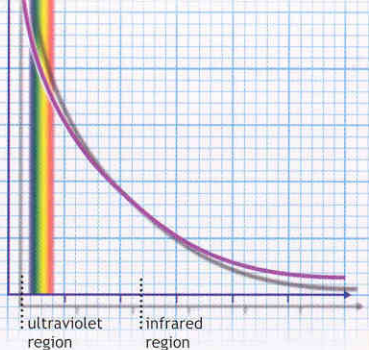
Confronted by this inexplicable phenomenon, in 1900 the German physicist Max Planck was reduced to the desperate remedy of solving the black body puzzle by assuming that light could be divided up into little chunks, pieces that he called "quanta," instead of always being a smooth, continuous wave. In order to do this, he assigned an energy, E , to each quantum that was related to its frequency, f .



Max Planck

(1858–1947) started the quantum revolution in 1900, when he explained

black body radiation in terms of light quanta. Planck never really came to terms with the revolution he had started, but was open-minded enough to be one of the first people to appreciate the importance of the Special Theory of Relativity when it was published in 1905. In 1918, he received the Nobel Prize for his scientific work.



predicted curve
This is the radiation curve that the classical physics of Newton and Maxwell predicted. Any hot object should radiate energy mainly at very short wavelengths, which is the ultraviolet end of the spectrum. The failure of this prediction has become known as the "ultraviolet catastrophe."